MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **SPECIFICATION**
39EJ DEMINERALISATION SYS.
DEMIN WATER TANK

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DEFINITIONS AND ABBREVIATIONS

Employer	MA'ADEN GOLD AND BASE METALS or their representatives
EPC Contractor	Consortium of Outotec (OT) and Larsen & Toubro (L&T)
Project management Company (PMC)	BECHTEL appointed by EMPLOYER
Supplier	An entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Sub- supplier	A sub entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Shall	Is to be understood as mandatory in relation to the requirements of this Document
Should	Is to be understood as a strong recommendation to comply with the requirements of this Document
TBC	To be confirmed
DN	Diametre Nominel
ASTM	American Society for Testing and Materials
ASME	American Society of Mechanical Engineers
API	American Petroleum Institute
EN	European Standard
RFSO	Raised-Face Slip On
N/A	Not Applicable
UoM	Unit of Measurement
Mt/a	Metric Tonnes Per Annum
P& ID	Piping and Instrumentation Diagram
No	Number
m	Metre
mm	Millimetre
ºC	Degree Centigrade
m_3/h	Cubic Meter Per Hour
kPa	Kilo Pascal
kPag	Kilo Pascal Gauge

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1. INTENT

The purpose of this document is to prepare a basic guideline for designing of the Demineralized Water Storage Tank for Demineralized Water System in 4.0 Mt/a Capacity Mansourah Massarah Gold Project at KSA.

2. SCOPE

The scope of work shall include design, engineering, purchasing, fabrication, manufacturing, testing, packing, forwarding to site, storing of material, erection, commissioning & site performance testing.

3. TECHNICAL SPECIFICATION

Equipment Number	39EJ-TNK-001

3.1. Tank Description

Name	Value	UoM
Tank Description	Demin Water Tank	
Location	Gold Processing Plant	
No. of Tanks	One (1)	No
Tank Contents	Demineralized Water	
Insulation Required	No	
Painting Required (External)	Yes	
Painting Required (Internal)	Yes	
Roof Approach	Staircase	
Corrosion Protection	No	

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3.2. Design Parameters

Description	Value	UoM
Design Outlet Flow Rate for each tank	137 (TBC)	m^3/h
Design Fill Rate for each tank	137 (TBC)	m^3/h
Roof Required	Yes	
Roof Type	Supported cone roof with maintenance personnel access	
Roof Supported/Unsupported	Supported on rafter/truss	
Anchoring	Yes	
Roof Uniform Live Load	1.2	kPa
Foundation Type	Ring wall	
Agitator/Rake	No	
Agitator Type	N/A	
Agitator/Rake supported by tank	N/A	
Structural Material Specification	ASTM A36 GR. 36 / EN 10025 GR. S275 JR / S355 JR	
Structural Bolting	ASTM A325	
Attachment/ Nozzle Bolting	ASTM A193 GR. B7 / ASTM A194 GR. 2H	
Plate Material Specification (Standard & Grade)	ASTM A36 GR. 36 or EN 10025 GR. S275 J0 / S355 J0	
Stress Relieved for Process Reasons	As per API 650	
Design Basic (Shell, Roof & Bottom)	API 650, 12th edition, Addendum 3, August 2018	
Flange Basic	ASME B16.5 ≤ DN 600 ASME B16.47 > DN 600	
Vent Design Basic	API 2000	

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Description	Value	UoM
Painting	External – Shell and Roof: Epoxy coat over Inorganic Zinc primer/ intermediate coats Bottom: Epoxy Ladder, stage, stage rolling ladder: Hot-Dip Galvanized (not painted)	

3.3. Process Parameters

Name	Value	UoM
Design Pressure	Atmospheric	kPag
Normal Operating Temperature	40	deg C
Maximum Operating Temperature	44	deg C
Effective Capacity	548	m^3
Minimum Operating Level	1010 TBC	mm
Maximum Operating Level	10640 TBC	mm
Maximum Density of Contents	1000	kg/m^3
Design Scale Thickness	0	mm
Density of Scale	0	kg/m^3

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3.4. Dimensional Parameters

Description	Value	UoM
Tank Diameter (I.D.)	8650	mm
Tank Cylindrical Height (excluding cone height)	12000	mm
Nominal capacity (at overflow level)	625 (TBC)	m^3
Floor Slope	1/120 conical apex-up	
Roof Slope	1 in 12	

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Description	Value	UoM
Corrosion Allowance Bottom	1.6	mm
Corrosion Allowance Roof	0	mm
Corrosion Allowance Shell	1.6	mm
Corrosion Allowance Structural members (element) supporting roof	1.6	mm
Liner Material	Natural Rubber Shore Hardness 65 ± 5 A	
Liner Thickness	6	mm
Insulation Material	N/A	
Insulation thickness - Roof	0	mm
Insulation Thickness - Shell	0	mm
Insulation Thickness - Bottom	0	mm

3.5. Nozzle Schedule

Purpose	Size (DN)	Qty.	Type	Location
Inlet – Demin water	150 (TBC)	1	150# RFSO	Shell
Demin Recirculation line	80 (TBC)	1	150# RFSO	Shell
Vent/ CO2 Breather	200 (TBC)	1	150# RFSO	Roof
Drain	100	1	150# RFSO	Shell
Outlet – Demin Water	200	2	150# RFSO	Shell
Overflow	250	1	150# RFSO	Shell
Clean Out Door (COD) – h x b	900 x 1200	1	API 650	Shell
Spare C/W Blind	100	1	150# RFSO	Roof
Spare C/W Blind	80	1	150# RFSO	Shell
Level Transmitter (LT)	100	1	150# RFSO	Roof
Manhole C/W Hinged Blind	600	1	API 650	Roof
Manhole C/W Davit	750	1	API 650	Shell

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REFERENCES

[1] PFD Services / Utilities – Demineralized Water; MD-216-39EJ-EG-PR-PFD-1001

[2] Design Criteria - Water System; MD-216-39EJ-EG-ME-CRT-1001

[3] Welding Guidelines; API 650 and MD-101-SMEM-EG-GE-SPC-0002-C01

[4] Mechanical Design Criteria - Stationary Equipment; MD-216-0000-EG-ME-CRT-0001-C01

[5] Metallic Storage Tanks; MD-101-SMEM-EG-ME-SPC-0013-C01

[6] Protective Coating; MD-101-SMEM-EG-ME-SPC-0007-C02

[7] Protective Coating of Structural Steelwork; MD-101-SMEM-EG-ST-SPC-0009-C01

[8] P&ID Services / Utilities – Demineralized Water; MD-216-39EJ-EG-PR-PID-1001

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